TSMC (2330.TW)

Earnings review: 3Q26 inline but 4Q26 GM came in slightly softer; AI demand outlook shows no signs of slowdown; reiterate Buy

2330.TW	12m Price Target: **NT$3,300.00**	Price: **NT$2,440.00**	Upside: **35.2%**
TSM	12m Price Target: **$660.00**	Price: **$430.00**	Upside: **53.5%**

BUY

Evelyn Yu
+886(2)2730-4187 | evelyn.yu@gs.com
Goldman Sachs (Asia) L.L.C., Taipei Branch

Allen Chang
+852-2978-2930 | allen.k.chang@gs.com
Goldman Sachs (Asia) L.L.C.

James Schneider, Ph.D.
+1(212)357-2929 | jim.schneider@gs.com
Goldman Sachs & Co. LLC

Stronger revenue momentum while GM guidance came in softer

TSMC hosted its 3Q26 analyst meeting on October 15. While we view the higher full-year 2026 revenue guidance and the further acceleration in capacity buildout as the key positives, the GM guidance could be the softer spot. In the meeting, management further raised its 2026 revenue growth outlook to grow by around 44% YoY (from slightly above 40%; USD term), supported by continued strong demand especially for AI/HPC. However, both 3Q26 GM result and 4Q26 GM guidance came in lower than our expectations; we believe this is to reflect a steeper N2 ramp and overseas fab dilution in the near term.

AI demand still ahead of supply, with no sign of slowdown

Key questions among investors largely surrounded demand outlook. Overall, we think that management's tone on AI demand remained very bullish. While TSMC did not formally update its long-term AI accelerator revenue CAGR, management indicated the CAGR will probably be higher, noting demand is extremely robust with no sign of slowdown and still far exceeds supply, with multiple drivers reinforcing this view, including - 1) agentic AI driving incremental CPU demand, which is excluded from TSMC's AI accelerator definition and implies upside beyond the headline CAGR; 2) broader adoption of open-source models lowering the barrier to AI usage and lifting leading-edge silicon demand; 3) continued demand for networking chips alongside AI accelerators; and 4) TSMC's own checks on US data center build progress and power supply, where management sees CSPs as very well prepared.

Capacity build-out accelerating again: N2 capacity more than double in 2027; additional N3 fab addition; Texas under evaluation

On N2, management for the first time guided its N2 capacity will be

Key Data

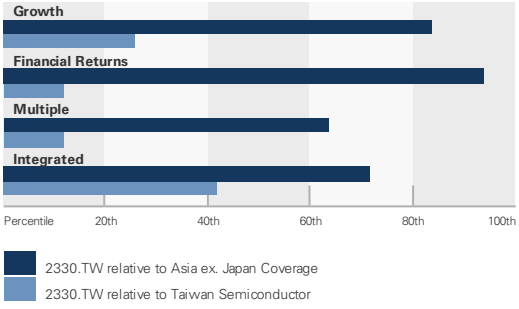
Market cap: NT$63.3tr / $2.0tr
Enterprise value: NT$60.4tr / $1.9tr
3m ADTV: NT$63.8bn / $2.0bn
Taiwan
Taiwan Semiconductor
M&A Rank: 3
Leases incl. in net debt & EV?: Yes

GS Forecast

	12/25	12/26E	12/27E	12/28E
Revenue (NT$ mn) New	**3,809,054.3**	**5,620,115.7**	**7,683,481.6**	**9,941,180.4**
Revenue (NT$ mn) Old	3,809,054.3	5,533,119.0	7,616,958.4	9,935,012.5
EBITDA (NT$ mn)	2,624,188.0	4,148,113.5	5,795,566.1	7,648,573.7
EPS (NT$) New	**66.26**	**112.35**	**150.01**	**196.90**
EPS (NT$) Old	66.26	110.77	150.00	195.67
P/E (X)	17.6	21.7	16.3	12.4
P/B (X)	5.6	8.2	5.8	4.2
Dividend yield (%)	1.9	1.2	1.4	1.5
CROCI (%)	24.3	34.1	39.5	42.6

	6/26	9/26E	12/26E	3/27E
EPS (NT$)	27.25	29.43	33.61	34.26

GS Factor Profile



Source: Company data, Goldman Sachs Research estimates.
See disclosures for details.

Goldman Sachs does and seeks to do business with companies covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision. For Reg AC certification and other important disclosures, see the Disclosure Appendix, or go to www.gs.com/research/hedge.html. Analysts employed by non-US affiliates are not registered/qualified as research analysts with FINRA in the U.S.

BUY

TSMC (2330.TW)

Rating since May 7, 2019

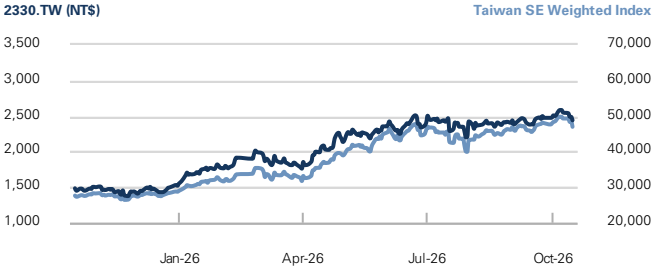
Ratios & Valuation

	12/25	12/26E	12/27E	12/28E
P/E (X)	17.6	21.7	16.3	12.4
P/B (X)	5.6	8.2	5.8	4.2
FCF yield (%)	3.3	2.6	3.9	5.8
EV/EBITDAR (X)	10.8	14.6	10.1	7.3
EV/EBITDA (excl. leases) (X)	10.8	14.6	10.1	7.3
CROCI (%)	24.3	34.1	39.5	42.6
ROE (%)	35.4	44.3	41.9	39.4
Net debt/equity (%)	(34.3)	(37.7)	(42.3)	(49.0)
Net debt/equity (excl. leases) (%)	(34.3)	(37.7)	(42.3)	(49.0)
Interest cover (X)	99.1	159.9	350.7	1,021.0
Days inventory outst, sales	27.6	24.8	25.7	24.5
Receivable days	26.6	27.4	27.6	24.6
Days payable outstanding	157.7	193.6	200.5	197.4
DuPont ROE (%)	31.5	37.5	35.7	33.8
Turnover (X)	0.5	0.5	0.6	0.5
Leverage (X)	1.5	1.4	1.3	1.2
Gross cash invested (ex cash) (NT$)	9,199,168.9	11,312,090.4	13,935,574.6	16,893,345.7
Average capital employed (NT$)	3,371,689.3	4,215,069.1	5,565,337.1	6,999,414.8
BVPS (NT$)	208.99	298.08	418.09	580.99

Growth & Margins (%)

	12/25	12/26E	12/27E	12/28E
Total revenue growth	31.6	47.5	36.7	29.4
EBITDA growth	32.2	58.1	39.7	32.0
EPS growth	46.4	69.6	33.5	31.3
DPS growth	29.4	31.8	13.8	12.1
EBIT margin	50.8	59.1	60.1	61.5
EBITDA margin	68.9	73.8	75.4	76.9
Net income margin	45.1	51.8	50.6	51.4

Price Performance



	3m	6m	12m
Absolute	0.0%	17.3%	66.6%
Rel. to the Taiwan SE Weighted Index	(2.8)%	(8.3)%	(3.3)%

Source: FactSet. Price as of 15 Oct 2026 close.

Income Statement (NT$ mn)

	12/25	12/26E	12/27E	12/28E
Total revenue	3,809,054.3	5,620,115.7	7,683,481.6	9,941,180.4
Cost of goods sold	(1,527,760.3)	(1,874,702.8)	(2,550,456.2)	(3,208,678.0)
SG&A	(98,775.0)	(103,546.5)	(124,628.0)	(142,028.0)
R&D	(246,427.3)	(321,860.8)	(387,916.0)	(479,916.0)
Other operating inc./(exp.)	–	–	–	–
EBITDA	**2,624,188.0**	**4,148,113.5**	**5,795,566.1**	**7,648,573.7**
Depreciation & amortization	(688,096.4)	(828,107.9)	(1,175,084.7)	(1,538,015.2)
EBIT	**1,936,091.7**	**3,320,005.6**	**4,620,481.4**	**6,110,558.5**
Net interest inc./(exp.)	86,206.5	91,497.7	97,520.8	104,678.2
Income/(loss) from associates	5,496.6	3,123.4	0.0	0.0
Pre-tax profit	**2,041,662.8**	**3,501,713.8**	**4,718,002.1**	**6,215,236.6**
Provision for taxes	(326,266.1)	(587,022.9)	(826,845.9)	(1,107,976.6)
Minority interest	2,485.8	(1,094.5)	(1,108.0)	(1,108.0)
Preferred dividends	–	–	–	–
Net inc. (pre-exceptionals)	**1,717,882.6**	**2,913,596.4**	**3,890,048.2**	**5,106,152.1**
Post-tax exceptionals	–	–	–	–
Net inc. (post-exceptionals)	**1,717,882.6**	**2,913,596.4**	**3,890,048.2**	**5,106,152.1**
EPS (basic, pre-except) (NT$)	**66.26**	**112.35**	**150.01**	**196.90**
EPS (diluted, pre-except) (NT$)	**66.26**	**112.35**	**150.01**	**196.90**
EPS (basic, post-except) (NT$)	**66.26**	**112.35**	**150.01**	**196.90**
EPS (diluted, post-except) (NT$)	**66.26**	**112.35**	**150.01**	**196.90**
DPS (NT$)	22.00	29.00	33.00	37.00
Div. payout ratio (%)	33.2	25.8	22.0	18.8

Balance Sheet (NT$ mn)

	12/25	12/26E	12/27E	12/28E
Cash & cash equivalents	2,767,856.4	3,498,381.5	4,868,723.0	7,480,069.9
Accounts receivable	282,059.2	560,521.2	600,545.0	737,328.5
Inventory	288,109.5	474,146.2	608,341.7	726,932.3
Other current assets	479,105.8	603,826.0	603,826.0	603,826.0
Total current assets	**3,817,130.8**	**5,136,874.8**	**6,681,435.7**	**9,548,156.6**
Net PP&E	3,691,840.9	5,028,894.0	6,565,521.7	8,152,618.9
Net intangibles	24,952.6	19,393.2	10,680.8	1,968.4
Total investments	172,370.4	227,935.7	227,935.7	227,935.7
Other long-term assets	226,729.2	255,063.0	255,063.0	255,063.0
Total assets	**7,933,023.9**	**10,668,160.7**	**13,740,636.8**	**18,185,742.6**
Accounts payable	714,546.9	1,274,278.1	1,528,013.1	1,942,016.1
Short-term debt	0.0	–	–	–
Short-term lease liabilities	–	–	–	–
Other current liabilities	743,472.4	875,205.6	875,205.6	875,205.6
Total current liabilities	**1,458,019.3**	**2,149,483.7**	**2,403,218.7**	**2,817,221.7**
Long-term debt	896,062.0	564,263.7	264,263.7	64,263.7
Long-term lease liabilities	–	–	–	–
Other long-term liabilities	118,147.3	179,158.2	179,158.2	179,158.2
Total long-term liabilities	**1,014,209.3**	**743,421.9**	**443,421.9**	**243,421.9**
Total liabilities	**2,472,228.6**	**2,892,905.6**	**2,846,640.6**	**3,060,643.6**
Preferred shares	–	–	–	–
Total common equity	5,419,596.0	7,729,951.3	10,841,990.2	15,066,390.7
Minority interest	**41,199.3**	**45,303.8**	**52,006.0**	**58,708.3**
Total liabilities & equity	**7,933,023.9**	**10,668,160.7**	**13,740,636.8**	**18,185,742.6**
Net debt, adjusted	(1,871,794.4)	(2,934,117.8)	(4,604,459.3)	(7,415,806.2)

Cash Flow (NT$ mn)

	12/25	12/26E	12/27E	12/28E
Net income	1,717,882.6	2,913,596.4	3,890,048.2	5,106,152.1
D&A add-back	688,096.4	828,107.9	1,175,084.7	1,538,015.2
Minority interest add-back	(2,485.8)	1,094.5	1,108.0	1,108.0
Net (inc)/dec working capital	98,731.8	95,232.4	79,515.7	158,628.9
Other operating cash flow	(227,249.3)	(171,751.0)	0.0	0.0
Cash flow from operations	**2,274,975.6**	**3,666,280.3**	**5,145,756.7**	**6,803,904.2**
Capital expenditures	(1,272,410.5)	(2,032,215.9)	(2,703,000.0)	(3,116,400.0)
Acquisitions	–	–	–	–
Divestitures	–	–	–	–
Others	128,017.1	(2,899.4)	–	–
Cash flow from investing	**(1,144,393.4)**	**(2,035,115.4)**	**(2,703,000.0)**	**(3,116,400.0)**
Repayment of lease liabilities	–	–	–	–
Dividends paid (common & pref)	(440,851.8)	(622,379.8)	(778,009.3)	(881,751.6)
Inc/(dec) in debt	40,448.1	(307,998.9)	(300,000.0)	(200,000.0)
Other financing cash flows	(89,949.1)	29,738.9	5,594.2	5,594.2
Cash flow from financing	**(490,352.9)**	**(900,639.9)**	**(1,072,415.1)**	**(1,076,157.3)**
Total cash flow	**640,229.4**	**730,525.1**	**1,370,341.6**	**2,611,346.8**
Free cash flow	1,002,565.1	1,634,064.4	2,442,756.7	3,687,504.2

Source: Company data, Goldman Sachs Research estimates.

more than double in 2027. As for N3, management announced an additional 3nm fab expansion in Tainan on strong customer demand, with volume production in 1H28. This comes on top of 1) the ongoing N3 fab expansion in Tainan (ramp in 1H27), 2) Arizona fab 2 (ramp in 3Q27, slightly ahead of its prior schedule), 3) Japan fab 2 (ramp in 2028), and 4) continued N5-to-N3 tool conversion. Beyond Arizona, TSMC confirmed it is evaluating Texas as a potential additional US site, and is in discussion with one customer building Terafab on utilizing that capacity; while the business model and structure remains under discussion, TSMC highlighted that its principle is to run the fab by itself.

GM: steeper N2 dilution near term, but "no ceiling" long term

3Q26 GM declined 1.3ppt QoQ to 66.4%, primarily on dilution from the initial N2 ramp, and 4Q26 GM is guided down a further 40bps to 66.0% at the mid-point on continued N2 dilution and overseas fabs. Management now expects overseas fab dilution to be closer to 2% for full-year 2026 (vs. 2-3% previously), helped by larger scale and cost improvement, but reiterated it will widen to 3-4% in the latter part of the five-year period as more US fabs ramp. Longer term, management stated there is "no ceiling" for TSMC's profitability, though any dramatic increase will not happen overnight. On pricing, management reiterated a strategic approach by segment, location and technology, with US production priced differently from Taiwan. Overall, we slightly lower our GM to 66.6%/66.8%/67.7% in 2026-2028E (vs. 67.2%/67.5%/67.8% previously).

CapEx: 2026 unchanged, multi-year step-up reiterated

2026 CapEx guidance was unchanged at US$60-64bn. While management did not provide numeric guidance beyond 2026, but reiterated that 2026-2028 CapEx will be significantly higher than the prior three years, and that revenue growth should continue to outpace CapEx growth. Management acknowledged the tools, material and labor inflation, but said the majority of CapEx is driven by long-term demand. Net net, we maintain our 2026E/2027E/2028E CapEx at US$64bn/US$85bn/US$98bn.

High confidence in its structural lead

Addressing questions regarding rising competition, management remains confident in its technology lead for both front end wafer and advanced packaging technology. Despite the later adoption timeframe (planned for in 2030) vs peers on High NA EUV, TSMC does not see the later adoption than peers as a disadvantage. In advanced packaging, TSMC is in volume production at 5x reticle size, which management believes no peer can match today, with 9x and 14x in development and CoPoS for future cost reduction. In addition, management also highlighted that it expects A14 (volume production in 2028) to be a bigger node than N2, with customer engagement already exceeding N2's at the same stage.

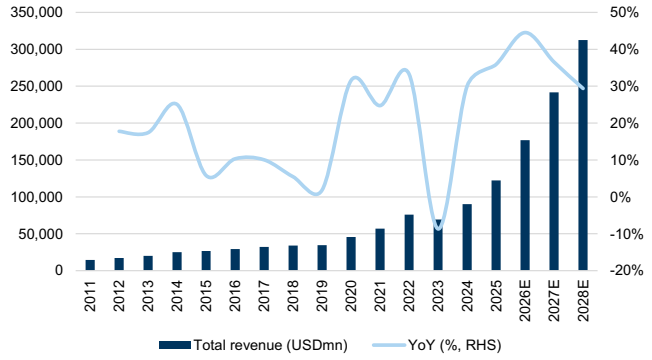
Key tables and charts

Exhibit 1: TSMC financial snapshot

TSMC (2330.TW)	2019	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E	25-28E CAGR
P&L (NT$bn)											
Revenue (US$bn)	34.7	45.6	56.9	75.9	69.3	90.1	122.4	177.0	241.6	312.6	36.7%
%YoY	1.9%	31.6%	24.7%	33.3%	-8.7%	30.0%	35.9%	44.5%	36.5%	29.4%	
Revenue	1,070	1,339	1,587	2,264	2,162	2,894	3,809	5,620	7,683	9,941	37.7%
%YoY	3.7%	25.2%	18.5%	42.6%	-4.5%	33.9%	31.6%	47.5%	36.7%	29.4%	
Gross profit	493	711	820	1,348	1,175	1,624	2,281	3,745	5,133	6,733	43.4%
%GM	46.0%	53.1%	51.6%	59.6%	54.4%	56.1%	59.9%	66.6%	66.8%	67.7%	
EBIT	373	567	650	1,121	921	1,322	1,936	3,320	4,620	6,111	46.7%
%OpM	34.8%	42.3%	40.9%	49.5%	42.6%	45.7%	50.8%	59.1%	60.1%	61.5%	
Net income	345	518	597	1,017	838	1,173	1,718	2,914	3,890	5,106	43.8%
%YoY	-1.7%	50.0%	15.2%	70.4%	-17.5%	39.9%	46.4%	69.6%	33.5%	31.3%	
EPS (NT$)	**13.3**	**20.0**	**23.0**	**39.2**	**32.3**	**45.3**	**66.3**	**112.4**	**150.0**	**196.9**	**43.8%**

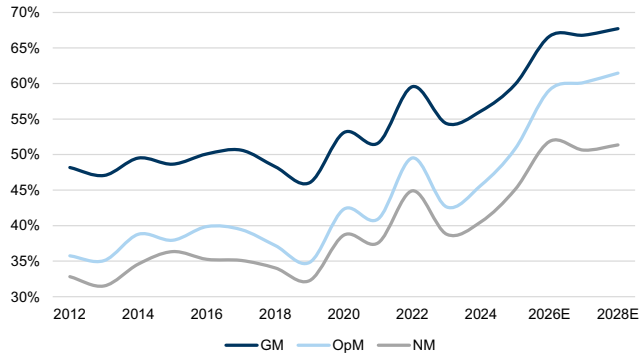
Source: Company data, Goldman Sachs Global Investment Research, Bloomberg

Exhibit 2: TSMC's revenue growth outlook



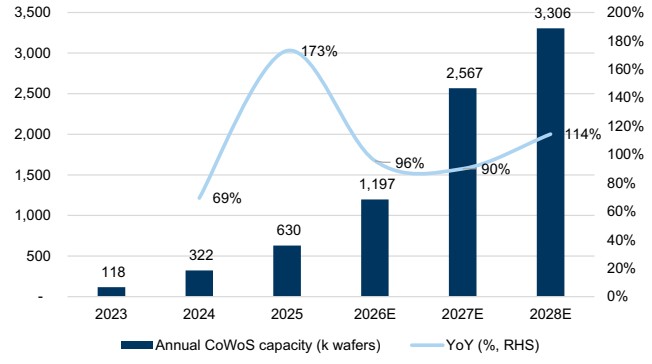
Source: Company data, Goldman Sachs Global Investment Research

Exhibit 3: TSMC's margin outlook



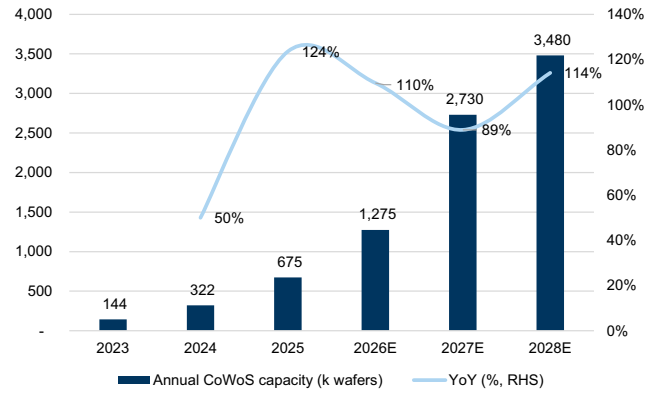
Source: Company data, Goldman Sachs Global Investment Research

Exhibit 4: TSMC's annual CoWoS shipment growth trend



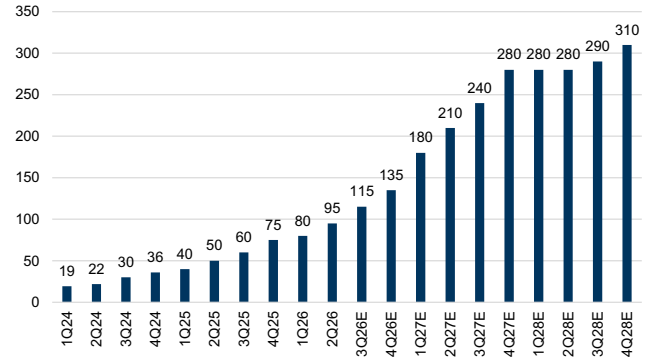
Source: Company data, Goldman Sachs Global Investment Research

Exhibit 5: TSMC's annual CoWoS capacity growth trend



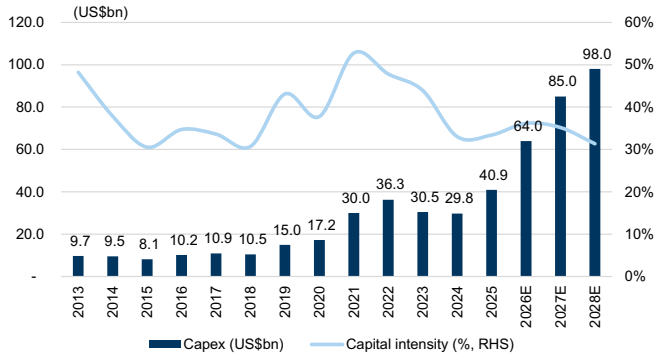
Source: Company data, Goldman Sachs Global Investment Research

Exhibit 6: TSMC's quarterly CoWoS capacity ramp (kwpm)



Source: Company data, Goldman Sachs Global Investment Research

Exhibit 7: TSMC's capex and capital intensity trend



Source: Company data, Goldman Sachs Global Investment Research

3Q26 results and 4Q26 guidance recap

4Q26 guidance: revenue well above expectations, GM below GSe

For 4Q26, management guided revenue to US$52.3-54.3bn (+14% QoQ at the mid-point; based on FX [TWD/USD] of 31.8), GM to 65-67% and OpM to 56.5-58.5%. Revenue guidance came in well above GSe of +11.0% QoQ, supported by continued strong leading-edge demand including the steep N2 ramp. However, the GM mid-point of 66.0% is 1.3ppt below GSe of 67.3% and broadly in line with consensus of 66.1%, as continued N2 dilution and overseas fabs are only partly offset by productivity gains and cross-node capacity optimization.

3Q26 results: revenue and EPS beat, GM below GSe

3Q26 revenue of NT$1,494.2bn (US$46.72bn) was up 17.6% QoQ and 50.9% YoY, above the high end of guidance in USD terms and above GSe/consensus by 0.8%/2.1%. EPS of NT$29.43 was up 8.0% QoQ and 68.7% YoY, above GSe/consensus by 1.2%/3.7%; excluding the NT$2.24 one-off gain from the VIS disposal in 2Q26, EPS grew 18% QoQ. GM of 66.4% (-1.3ppt QoQ) landed in the upper half of guidance (65-67%) but 1.1ppt below GSe, while OpM of 58.7% beat the high end of guidance (56-58%) on stable OpEx at 7.8% of revenue.

By technology, N2 contributed 8% of wafer revenue (vs. 3% in 2Q26) and N3 33% (vs. 30%), taking advanced nodes (7nm and below) to 79% (vs. 77%). By platform, HPC rose 22% QoQ to 68% of revenue (vs. 66%), with smartphone +10% QoQ at 21%. Inventory days fell 5 days QoQ to 82 days on higher wafer shipments.

Exhibit 8: 3Q26 results summary

	3Q26	2Q26	3Q25	%QoQ	%YoY	GS prev.	%Diff	Consensus	%Diff
(NT$mn)									
Revenue	1,494,243	1,270,380	989,918	17.6%	50.9%	1,482,698	0.8%	1,463,320	2.1%
Gross profit	992,379	860,311	588,543	15.4%	68.6%	1,000,438	-0.8%	972,771	2.0%
Op. income	876,445	766,603	500,685	14.3%	75.0%	892,956	-1.8%	854,354	2.6%
Net income	763,064	706,562	452,301	8.0%	68.7%	753,824	1.2%	738,565	3.3%
EPS (NT$)	29.43	27.25	17.44	8.0%	68.7%	29.07	1.2%	28.38	3.7%
GM	66.4%	67.7%	59.5%	-1.3%	7.0%	67.5%	-1.1%	66.5%	-0.1%
OpM	58.7%	60.3%	50.6%	-1.7%	8.1%	60.2%	-1.6%	58.4%	0.5%
NM	51.1%	55.6%	45.7%	-4.6%	5.4%	50.8%	0.2%	50.5%	1.2%

Source: Bloomberg, Company data, Goldman Sachs Global Investment Research

Earnings revisions, valuation and key risks

Forecast changes

Post its earnings results, we tweak our 2026/2027/2028E EPS by 1.4%/0.0%/0.6% to reflect: 1) stronger 4Q26 revenue guidance and the higher 2026 growth outlook, but partly offset by 2) lower GM assumptions on steeper N2 ramp and overseas fab dilution.

Exhibit 9: Earnings revisions

	2026E			2027E			2028E		
(NT$mn)	Old	New	Diff (%)	Old	New	Diff (%)	Old	New	Diff (%)
Revenue	5,533,119	5,620,116	1.6%	7,616,958	7,683,482	0.9%	9,935,012	9,941,180	0.1%
Gross profit	3,720,174	3,745,413	0.7%	5,140,114	5,133,025	-0.1%	6,739,653	6,732,502	-0.1%
Op. income	3,314,172	3,320,006	0.2%	4,645,886	4,620,481	-0.5%	6,105,525	6,110,558	0.1%
Net income	2,872,580	2,913,596	1.4%	3,889,923	3,890,048	0.0%	5,074,175	5,106,152	0.6%
EPS (NT$)	110.77	112.35	1.4%	150.00	150.01	0.0%	195.67	196.90	0.6%
GM	67.2%	66.6%	-0.6%	67.5%	66.8%	-0.7%	67.8%	67.7%	-0.1%
OpM	59.9%	59.1%	-0.8%	61.0%	60.1%	-0.9%	61.5%	61.5%	0.0%
NM	51.9%	51.8%	-0.1%	51.1%	50.6%	-0.4%	51.1%	51.4%	0.3%

	4Q26E			1Q27E			2Q27E		
(NT$mn)	Old	New	Diff (%)	Old	New	Diff (%)	Old	New	Diff (%)
Revenue	1,645,937	1,721,389	4.6%	1,711,413	1,778,044	3.9%	1,846,140	1,865,007	1.0%
Gross profit	1,108,130	1,141,428	3.0%	1,155,117	1,178,067	2.0%	1,248,183	1,245,106	-0.2%
Op. income	995,648	1,017,992	2.2%	1,040,535	1,053,531	1.2%	1,128,701	1,118,170	-0.9%
Net income	839,714	871,491	3.8%	880,339	888,340	0.9%	938,351	926,864	-1.2%
EPS (NT$)	32.38	33.61	3.8%	33.95	34.26	0.9%	36.18	35.74	-1.2%
GM	67.3%	66.3%	-1.0%	67.5%	66.3%	-1.2%	67.6%	66.8%	-0.8%
OpM	60.5%	59.1%	-1.4%	60.8%	59.3%	-1.5%	61.1%	60.0%	-1.2%
NM	51.0%	50.6%	-0.4%	51.4%	50.0%	-1.5%	50.8%	49.7%	-1.1%

Source: Goldman Sachs Global Investment Research

Exhibit 10: TSMC's P&L summary

	1Q26	2Q26	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2026E	2027E	2028E
P&L (NT$mn)											
Revenue	1,134,103	1,270,380	1,494,243	1,721,389	1,778,044	1,865,007	1,954,437	2,085,994	5,620,116	7,683,482	9,941,180
Gross profit	751,295	860,311	992,379	1,141,428	1,178,067	1,245,106	1,310,056	1,399,797	3,745,413	5,133,025	6,732,502
Operating profit	658,966	766,603	876,445	1,017,992	1,053,531	1,118,170	1,180,720	1,268,061	3,320,006	4,620,481	6,110,558
Net income	572,480	706,562	763,064	871,491	888,340	926,864	1,000,360	1,074,484	2,913,596	3,890,048	5,106,152
EPS (NT$)	22.08	27.25	29.43	33.61	34.26	35.74	38.58	41.43	112.35	150.01	196.90
Margins (%)											
Gross margin	66.2%	67.7%	66.4%	66.3%	66.3%	66.8%	67.0%	67.1%	66.6%	66.8%	67.7%
Operating profit margin	58.1%	60.3%	58.7%	59.1%	59.3%	60.0%	60.4%	60.8%	59.1%	60.1%	61.5%
Net margin	50.5%	55.6%	51.1%	50.6%	50.0%	49.7%	51.2%	51.5%	51.8%	50.6%	51.4%
YoY (%)											
Revenue	35.1%	36.0%	50.9%	64.6%	56.8%	46.8%	30.8%	21.2%	47.5%	36.7%	29.4%
Gross profit	52.3%	57.2%	68.6%	75.1%	56.8%	44.7%	32.0%	22.6%	64.2%	37.0%	31.2%
Operating profit	61.9%	65.4%	75.0%	80.2%	59.9%	45.9%	34.7%	24.6%	71.5%	39.2%	32.2%
Net income	58.3%	77.4%	68.7%	72.3%	55.2%	31.2%	31.1%	23.3%	69.6%	33.5%	31.3%
EPS	58.3%	77.4%	68.7%	72.3%	55.1%	31.2%	31.1%	23.3%	69.6%	33.5%	31.3%
QoQ (%)											
Revenue	8.4%	12.0%	17.6%	15.2%	3.3%	4.9%	4.8%	6.7%			
Gross profit	15.2%	14.5%	15.4%	15.0%	3.2%	5.7%	5.2%	6.9%			
Operating profit	16.7%	16.3%	14.3%	16.2%	3.5%	6.1%	5.6%	7.4%			
Net income	13.2%	23.4%	8.0%	14.2%	1.9%	4.3%	7.9%	7.4%			
EPS	13.2%	23.4%	8.0%	14.2%	1.9%	4.3%	7.9%	7.4%			

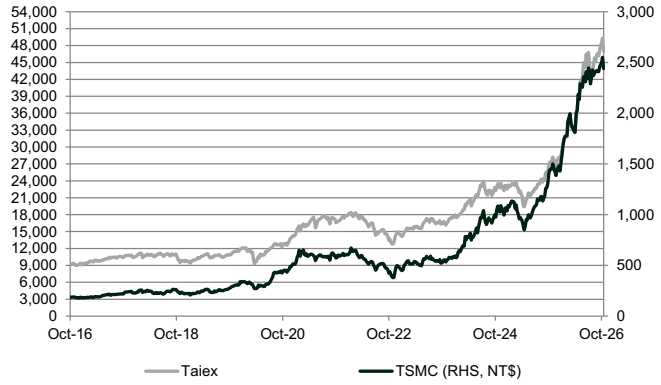
Source: Company data, Goldman Sachs Global Investment Research

Reiterate Buy; 12m TP maintain at NT$3,300

We maintain our 12m TP at NT$3,300 based on a target P/E of 22x (unchanged, benchmarked to 1.0stdv above 5-year avg. forward P/E) applied to our 2027E EPS. Accordingly, we maintain our 12 month TP for the ADR at US$660, based on an unchanged FX conversion of USD/TWD of 30.0 and unchanged ADR premium of 20%. Our 22x target P/E multiple is also benchmarked to TSMC's trading average of 22x during its previous revenue upcycle where its 3-year revenue CAGR (2019-2021) was 28%, vs. our 3-year revenue CAGR at 38% during 2025-2028E.

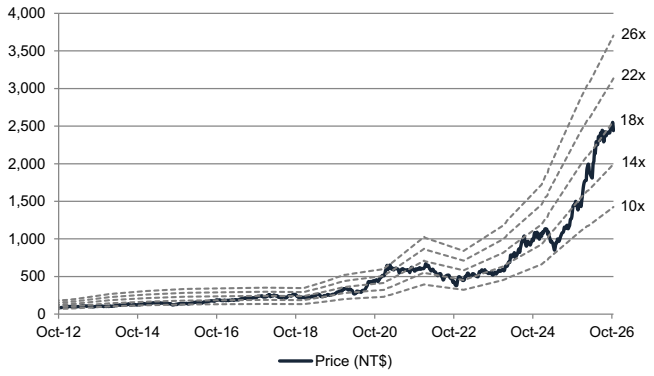
Our TP implies 35% (ADR 54%) upside to the last close. As a result, we reiterate our Buy ratings for TSMC/ADR. We maintain our constructive view on TSMC's long-term growth outlook as we believe the company's solid technology leadership and execution better positions it vs. peers to capture the industry's long-term structural growth opportunities, especially in AI.

Exhibit 11: Taiex vs. TSMC



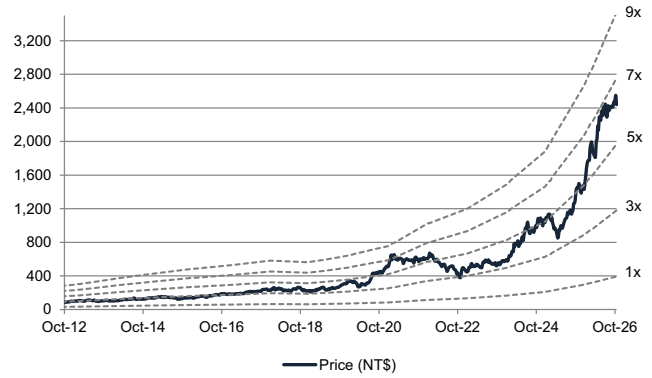
Source: Bloomberg

Exhibit 12: Forward P/E



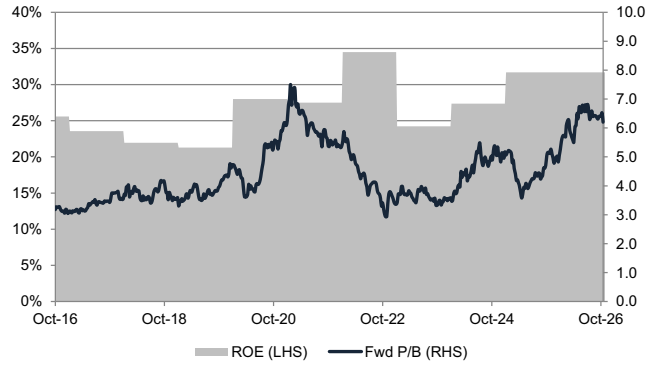
Source: Bloomberg

Exhibit 13: Forward P/B



Source: Bloomberg

Exhibit 14: P/B vs. ROE



Source: Bloomberg, Data compiled by Goldman Sachs Global Investment Research

Investment Thesis, Price Target Risks and Methodology

Investment Thesis - TSMC (2330.TW/TSM)

TSMC is a leading global foundry company specializing in cutting-edge nodes. Its leading technology stance enables it to enjoy more than 60% revenue share in the global foundry market. We like TSMC as we believe its solid technology leadership and execution better position it vs. peers to capture the industry's long-term structural growth opportunities, particularly in areas such as AI/5G/HPC/EV. In addition, valuation looks attractive in our view, with the shares trading at the mid-range of their 10-year trading history (P/E: 10-29x). Furthermore, we view TSMC as the key AI enabler among our Taiwan Semis coverage, thanks to its leadership stance in leading edge nodes and advanced packaging technology – CoWoS (chip on wafer on substrate). We believe TSMC will achieve its 25% revenue CAGR target for the next several years, driven primarily by increasing silicon content growth and AI/HPC demand, with LT GM to remain at 56%+. We are Buy rated.

Price Target Risks and Methodology - TSMC (2330.TW/TSM)

Valuation methodology: We have a 12m TP of NT$3,300, which is derived by applying a target P/E multiple of 22x to our 2027E EPS. Our 22x target P/E is benchmarked against 1.0stdv above its 5-year trading average. For the ADR (TSM), we have a 12m TP of US$660, based on a USD/TWD rate of 30.0 and an ADR premium of 20%.

Key downside risks to our views: (1) further deterioration in end-demand recovery impacting capacity utilization; (2) slower customer node migrations; (3) slowdown in AI investment resulting in lower long-term semiconductor content growth; (4) poor yields/execution resulting in worse-than-expected profitability; (5) stronger competition resulting in ASP/profitability erosion; and (6) unfavorable FX trend or higher-than-expected cost increase weighing on the margin outlook.

Disclosure Appendix

Reg AC

We, Evelyn Yu, Allen Chang and James Schneider, Ph.D., hereby certify that all of the views expressed in this report accurately reflect our personal views about the subject company or companies and its or their securities. We also certify that no part of our compensation was, is or will be, directly or indirectly, related to the specific recommendations or views expressed in this report.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Contributing Authors: Evelyn Yu Goldman Sachs (Asia) L.L.C., Taipei Branch, Allen Chang Goldman Sachs (Asia) L.L.C., James Schneider, Ph.D. Goldman Sachs & Co. LLC.

Unless otherwise stated, the individuals listed in the Contributing Authors disclosure of this report are analysts in Goldman Sachs' Global Investment Research division.

GS Factor Profile

The Goldman Sachs Factor Profile provides investment context for a stock by comparing key attributes to the market (i.e. our universe of rated stocks) and its sector peers. The four key attributes depicted are: Growth, Financial Returns, Multiple (e.g. valuation) and Integrated (a composite of Growth, Financial Returns and Multiple). Growth, Financial Returns and Multiple are calculated by using normalized ranks for specific metrics for each stock. The normalized ranks for the metrics are then averaged and converted into percentiles for the relevant attribute. The precise calculation of each metric may vary depending on the fiscal year, industry and region, but the standard approach is as follows:

Growth is based on a stock's forward-looking sales growth, EBITDA growth and EPS growth (for financial stocks, only EPS and sales growth), with a higher percentile indicating a higher growth company. **Financial Returns** is based on a stock's forward-looking ROE, ROCE and CROCI (for financial stocks, only ROE), with a higher percentile indicating a company with higher financial returns. **Multiple** is based on a stock's forward-looking P/E, P/B, price/dividend (P/D), EV/EBITDA, EV/FCF and EV/Debt Adjusted Cash Flow (DACF) (for financial stocks, only P/E, P/B and P/D), with a higher percentile indicating a stock trading at a higher multiple. The **Integrated** percentile is calculated as the average of the Growth percentile, Financial Returns percentile and (100% - Multiple percentile).

Financial Returns and Multiple use the Goldman Sachs analyst forecasts at the fiscal year-end at least three quarters in the future. Growth uses inputs for the fiscal year at least seven quarters in the future compared with the year at least three quarters in the future (on a per-share basis for all metrics).

For a more detailed description of how we calculate the GS Factor Profile, please contact your GS representative.

M&A Rank

Across our global coverage, we examine stocks using an M&A framework, considering both qualitative factors and quantitative factors (which may vary across sectors and regions) to incorporate the potential that certain companies could be acquired. We then assign a M&A rank as a means of scoring companies under our rated coverage from 1 to 3, with 1 representing high (30%-50%) probability of the company becoming an acquisition target, 2 representing medium (15%-30%) probability and 3 representing low (0%-15%) probability. For companies ranked 1 or 2, in line with our standard departmental guidelines we incorporate an M&A component into our target price. M&A rank of 3 is considered immaterial and therefore does not factor into our price target, and may or may not be discussed in research.

Quantum

Quantum is Goldman Sachs' proprietary database providing access to detailed financial statement histories, forecasts and ratios. It can be used for in-depth analysis of a single company, or to make comparisons between companies in different sectors and markets.

Disclosures

The rating(s) for TSMC and TSMC (ADR) is/are relative to the other companies in its/their coverage universe: All Ring Tech Co., Aspeed, Grand Plastic Technology Corp., Hon Precision, MPI Corp., MediaTek, TSMC, TSMC (ADR), Winway Technology Co.

Company-specific regulatory disclosures

The following disclosures relate to relationships between The Goldman Sachs Group, Inc. (with its affiliates, "Goldman Sachs") and companies covered by Goldman Sachs Global Investment Research and referred to in this research.

Goldman Sachs has received compensation for investment banking services in the past 12 months: TSMC (NT$2,440.00) and TSMC (ADR) ($430.00)

Goldman Sachs expects to receive or intends to seek compensation for investment banking services in the next 3 months: TSMC (NT$2,440.00) and TSMC (ADR) ($430.00)

Goldman Sachs has received compensation for non-investment banking services during the past 12 months: TSMC (NT$2,440.00) and TSMC (ADR) ($430.00)

Goldman Sachs had an investment banking services client relationship during the past 12 months with: TSMC (NT$2,440.00) and TSMC (ADR) ($430.00)

Goldman Sachs had a non-investment banking securities-related services client relationship during the past 12 months with: TSMC (NT$2,440.00) and TSMC (ADR) ($430.00)

Goldman Sachs had a non-securities services client relationship during the past 12 months with: TSMC (NT$2,440.00) and TSMC (ADR) ($430.00)

Goldman Sachs is a registered market maker in the securities or derivatives thereof: TSMC (NT$2,440.00) and TSMC (ADR) ($430.00)

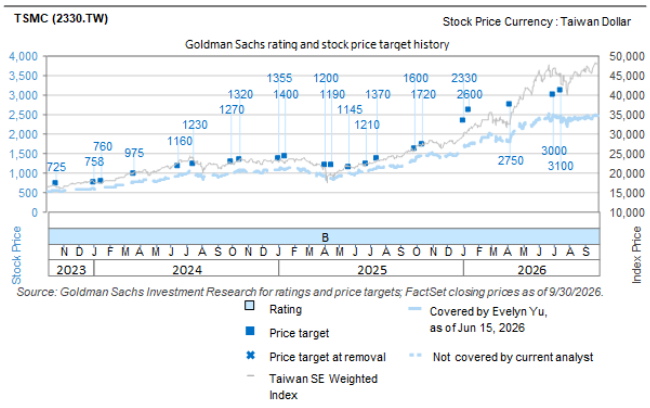
Distribution of ratings/investment banking relationships

Goldman Sachs Investment Research global Equity coverage universe

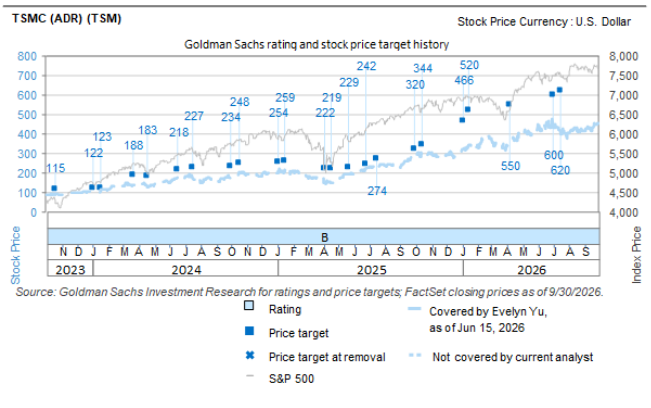
	Rating Distribution			Investment Banking Relationships		
	Buy	Hold	Sell	Buy	Hold	Sell
Global	50%	34%	16%	65%	58%	43%

As of October 1, 2026, Goldman Sachs Global Investment Research had investment ratings on 3,094 equity securities. Goldman Sachs assigns stocks as Buys and Sells on various regional Investment Lists; stocks not so assigned are deemed Neutral. Such assignments equate to Buy, Hold and Sell for the purposes of the above disclosure required by the FINRA Rules. See 'Ratings, Coverage universe and related definitions' below. The Investment Banking Relationships chart reflects the percentage of subject companies within each rating category for whom Goldman Sachs has provided investment banking services within the previous twelve months.

Price target and rating history chart(s)



The price targets shown should be considered in the context of all prior published Goldman Sachs research, which may or may not have included price targets, as well as developments relating to the company, its industry and financial markets.



The price targets shown should be considered in the context of all prior published Goldman Sachs research, which may or may not have included price targets, as well as developments relating to the company, its industry and financial markets.

Target price history table(s)

TSMC (2330.TW)

Date of report	Target price (NT$)	Closing price (NT$)
05-Oct-26	3,300.00	2,575.00
16-Jul-26	3,100.00	2,470.00
02-Jul-26	3,000.00	2,465.00
07-Apr-26	2,750.00	1,860.00
15-Jan-26	2,600.00	1,690.00
04-Jan-26	2,330.00	1,585.00
16-Oct-25	1,720.00	1,485.00
30-Sep-25	1,600.00	1,305.00
17-Jul-25	1,370.00	1,130.00
25-Jun-25	1,210.00	1,070.00
22-May-25	1,145.00	982.00
17-Apr-25	1,190.00	847.00
06-Apr-25	1,200.00	942.00
16-Jan-25	1,400.00	1,105.00
03-Jan-25	1,355.00	1,075.00
17-Oct-24	1,320.00	1,035.00
30-Sep-24	1,270.00	957.00
18-Jul-24	1,230.00	1,005.00
18-Jun-24	1,160.00	943.00
21-Mar-24	975.00	784.00
18-Jan-24	760.00	588.00
02-Jan-24	758.00	593.00
19-Oct-23	725.00	546.00

TSMC (ADR) (TSM)

Date of report	Target price ($)	Closing price ($)
05-Oct-26	660.00	485.80
16-Jul-26	620.00	409.74
02-Jul-26	600.00	434.16
07-Apr-26	550.00	345.32
15-Jan-26	520.00	341.64
04-Jan-26	466.00	319.61
16-Oct-25	344.00	299.84
30-Sep-25	320.00	279.29
17-Jul-25	274.00	245.60
25-Jun-25	242.00	222.74
22-May-25	229.00	196.19
17-Apr-25	219.00	151.74
06-Apr-25	222.00	146.80
16-Jan-25	259.00	214.79
03-Jan-25	254.00	208.61
17-Oct-24	248.00	205.84
30-Sep-24	234.00	173.67
18-Jul-24	227.00	171.87
18-Jun-24	218.00	179.69
18-Apr-24	183.00	132.27
21-Mar-24	188.00	139.45
18-Jan-24	123.00	113.03
02-Jan-24	122.00	101.53
19-Oct-23	115.00	92.91

Price targets shown in table(s) are unadjusted for corporate actions.

Regulatory disclosures

Disclosures required by United States laws and regulations

See company-specific regulatory disclosures above for any of the following disclosures required as to companies referred to in this report: manager or co-manager in a pending transaction; 1% or other ownership; compensation for certain services; types of client relationships; managed/co-managed public offerings in prior periods; directorships; registered market maker. Goldman Sachs trades or may trade as a principal in debt securities (or in related derivatives) of issuers discussed in this report.

The following are additional required disclosures: **Ownership and material conflicts of interest:** Goldman Sachs policy prohibits its analysts, professionals reporting to analysts and members of their households from owning securities of any company in the analyst's area of coverage. **Analyst**

compensation: Analysts are paid in part based on the profitability of Goldman Sachs, which includes investment banking revenues. **Analyst as officer or director:** Goldman Sachs policy generally prohibits its analysts, persons reporting to analysts or members of their households from serving as an officer, director or advisor of any company in the analyst's area of coverage. **Non-U.S. Analysts:** Non-U.S. analysts may not be associated persons of Goldman Sachs & Co. LLC and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with a subject company, public appearances and trading in securities covered by the analysts.

Additional disclosures required under the laws and regulations of jurisdictions other than the United States

The following disclosures are those required by the jurisdiction indicated, except to the extent already made above pursuant to United States laws and regulations. **Australia:** Goldman Sachs Australia Pty Ltd and its affiliates are not authorised deposit-taking institutions (as that term is defined in the Banking Act 1959 (Cth)) in Australia and do not provide banking services, nor carry on a banking business, in Australia. This research, and any access to it, is intended only for "wholesale clients" within the meaning of the Australian Corporations Act, unless otherwise agreed by Goldman Sachs. In producing research reports, members of Global Investment Research of Goldman Sachs Australia may attend site visits and other meetings hosted by the companies and other entities which are the subject of its research reports. In some instances the costs of such site visits or meetings may be met in part or in whole by the issuers concerned if Goldman Sachs Australia considers it is appropriate and reasonable in the specific circumstances relating to the site visit or meeting. To the extent that the contents of this document contains any financial product advice, it is general advice only and has been prepared by Goldman Sachs without taking into account a client's objectives, financial situation or needs. A client should, before acting on any such advice, consider the appropriateness of the advice having regard to the client's own objectives, financial situation and needs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests and a copy of Goldman Sachs' Australian Sell-Side Research Independence Policy Statement are available at: https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html. **Brazil:** Disclosure information in relation to CVM Resolution n. 20 is available at https://www.gs.com/worldwide/brazil/area/gir/index.html. Where applicable, the Brazil-registered analyst primarily responsible for the content of this research report, as defined in Article 20 of CVM Resolution n. 20, is the first author named at the beginning of this report, unless indicated otherwise at the end of the text. **Canada:** This information is being provided to you for information purposes only and is not, and under no circumstances should be construed as, an advertisement, offering or solicitation by Goldman Sachs & Co. LLC for purchasers of securities in Canada to trade in any Canadian security. Goldman Sachs & Co. LLC is not registered as a dealer in any jurisdiction in Canada under applicable Canadian securities laws and generally is not permitted to trade in Canadian securities and may be prohibited from selling certain securities and products in certain jurisdictions in Canada. If you wish to trade in any Canadian securities or other products in Canada please contact Goldman Sachs Canada Inc., an affiliate of The Goldman Sachs Group Inc., or another registered Canadian dealer. **Hong Kong:** Further information on the securities of covered companies referred to in this research may be obtained on request from Goldman Sachs (Asia) L.L.C. **India:** Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (India) Securities Private Limited, Research Analyst - SEBI Registration Number INH000001493, 10th Floor, Ascent-Worli, Sudam Kalu Ahire Marg, Worli, Mumbai-400 025, India, Corporate Identity Number U74140MH2006FTC160634, Phone +91 22 6616 9000, Fax +91 22 6616 9001. Goldman Sachs may beneficially own 1% or more of the securities (as such term is defined in clause 2 (h) the Indian Securities Contracts (Regulation) Act, 1956) of the subject company or companies referred to in this research report. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Goldman Sachs (India) Securities Private Limited compliance officer and investor grievance contact details, a copy of the annual compliance audit report and other relevant information and disclosures can be found at this link: https://www.goldmansachs.com/worldwide/india/research-analyst. **Japan:** See below. **Korea:** This research, and any access to it, is intended only for "professional investors" within the meaning of the Financial Services and Capital Markets Act, unless otherwise agreed by Goldman Sachs. Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (Asia) L.L.C., Seoul Branch. **New Zealand:** Goldman Sachs New Zealand Limited and its affiliates are neither "registered banks" nor "deposit takers" (as defined in the Reserve Bank of New Zealand Act 1989) in New Zealand. This research, and any access to it, is intended for "wholesale clients" (as defined in the Financial Advisers Act 2008) unless otherwise agreed by Goldman Sachs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests is available at: https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html. **Russia:** Research reports distributed in the Russian Federation are not advertising as defined in the Russian legislation, but are information and analysis not having product promotion as their main purpose and do not provide appraisal within the meaning of the Russian legislation on appraisal activity. Research reports do not constitute a personalized investment recommendation as defined in Russian laws and regulations, are not addressed to a specific client, and are prepared without analyzing the financial circumstances, investment profiles or risk profiles of clients. Goldman Sachs assumes no responsibility for any investment decisions that may be taken by a client or any other person based on this research report. **Singapore:** Goldman Sachs (Singapore) Pte. (Company Number: 198602165W), which is regulated by the Monetary Authority of Singapore, accepts legal responsibility for this research, and should be contacted with respect to any matters arising from, or in connection with, this research. **Taiwan:** This material is for reference only and must not be reprinted without permission. Investors should carefully consider their own investment risk. Investment results are the responsibility of the individual investor. **United Kingdom:** Persons who would be categorized as retail clients in the United Kingdom, as such term is defined in the rules of the Financial Conduct Authority, should read this research in conjunction with prior Goldman Sachs research on the covered companies referred to herein and should refer to the risk warnings that have been sent to them by Goldman Sachs International. A copy of these risks warnings, and a glossary of certain financial terms used in this report, are available from Goldman Sachs International on request.

European Union and United Kingdom: Disclosure information in relation to Article 6 (2) of the European Commission Delegated Regulation (EU) (2016/958) supplementing Regulation (EU) No 596/2014 of the European Parliament and of the Council (including as that Delegated Regulation is implemented into United Kingdom domestic law and regulation following the United Kingdom's departure from the European Union and the European Economic Area) with regard to regulatory technical standards for the technical arrangements for objective presentation of investment recommendations or other information recommending or suggesting an investment strategy and for disclosure of particular interests or indications of conflicts of interest is available at https://www.gs.com/disclosures/europeanpolicy.html which states the European Policy for Managing Conflicts of Interest in Connection with Investment Research.

Japan: Goldman Sachs Japan Co., Ltd. is a Financial Instrument Dealer registered with the Kanto Financial Bureau under registration number Kinsho 69, and a member of Japan Securities Dealers Association, Financial Futures Association of Japan Type II Financial Instruments Firms Association, and Investment Management Association of Japan. Sales and purchase of equities are subject to commission pre-determined with clients plus consumption tax. See company-specific disclosures as to any applicable disclosures required by Japanese stock exchanges, the Japanese Securities Dealers Association or the Japanese Securities Finance Company.

Ratings, coverage universe and related definitions

Buy (B), Neutral (N), Sell (S) Analysts recommend stocks as Buys or Sells for inclusion on various regional Investment Lists. Being assigned a Buy or Sell on an Investment List is determined by a stock's total return potential relative to its coverage universe. Any stock not assigned as a Buy or a Sell on an Investment List with an active rating (i.e., a stock that is not Rating Suspended, Not Rated, Early-Stage Biotech, Coverage Suspended or Not Covered), is deemed Neutral. Each region manages Regional Conviction Lists, which are selected from Buy rated stocks on the respective region's Investment Lists and represent investment recommendations focused on the size of the total return potential and/or the likelihood of the realization of the return across their respective areas of coverage. The addition or removal of stocks from such Conviction Lists are managed by the Investment Review Committee or other designated committee in each respective region and do not represent a change in the analysts' investment rating for such stocks.

Total return potential represents the upside or downside differential between the current share price and the price target, including all paid or anticipated dividends, expected during the time horizon associated with the price target. Price targets are required for all covered stocks. The total

return potential, price target and associated time horizon are stated in each report adding or reiterating an Investment List membership.

Coverage Universe: A list of all stocks in each coverage universe is available by primary analyst, stock and coverage universe at https://www.gs.com/research/hedge.html.

Not Rated (NR). The investment rating, target price and earnings estimates (where relevant) are removed pursuant to Goldman Sachs policy when Goldman Sachs is acting in an advisory capacity in a merger or in a strategic transaction involving this company, when there are legal, regulatory or policy constraints due to Goldman Sachs' involvement in a transaction, and in certain other circumstances. **Early-Stage Biotech (ES).** An investment rating and a target price are not assigned pursuant to Goldman Sachs policy when this company has neither a drug, treatment or medical device that has passed a Phase II clinical trial nor a license to distribute a post-Phase II drug, treatment or medical device. **Rating Suspended (RS).** Goldman Sachs Research has suspended the investment rating and price target for this stock, because there is not a sufficient fundamental basis for determining an investment rating or target price. The previous investment rating and target price, if any, are no longer in effect for this stock and should not be relied upon. **Coverage Suspended (CS).** Goldman Sachs has suspended coverage of this company. **Not Covered (NC).** Goldman Sachs does not cover this company.

Global product; distributing entities

Goldman Sachs Global Investment Research produces and distributes research products for clients of Goldman Sachs on a global basis. Analysts based in Goldman Sachs offices around the world produce research on industries and companies, and research on macroeconomics, currencies, commodities and portfolio strategy. This research is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Brazil by Goldman Sachs do Brasil Corretora de Títulos e Valores Mobiliários S.A.; Public Communication Channel Goldman Sachs Brazil: 0800 727 5764 and / or contatogoldmanbrasil@gs.com. Available Weekdays (except holidays), from 9am to 6pm. Canal de Comunicação com o Público Goldman Sachs Brasil: 0800 727 5764 e/ou contatogoldmanbrasil@gs.com. Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h; in Canada by Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs (Asia) L.L.C.; in India by Goldman Sachs (India) Securities Private Ltd.; in Japan by Goldman Sachs Japan Co., Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Russia by OOO Goldman Sachs; in Singapore by Goldman Sachs (Singapore) Pte. (Company Number: 198602165W); and in the United States of America by Goldman Sachs & Co. LLC. Goldman Sachs International has approved this research in connection with its distribution in the United Kingdom.

Goldman Sachs International ("GSI"), authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA, has approved this research in connection with its distribution in the United Kingdom.

European Economic Area: Goldman Sachs Bank Europe SE ("GSBE") is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and Deutsche Bundesbank and disseminates research within the European Economic Area.

General disclosures

This research is for our clients only. Other than disclosures relating to Goldman Sachs, this research is based on current public information that we consider reliable, but we do not represent it is accurate or complete, and it should not be relied on as such. The information, opinions, estimates and forecasts contained herein are as of the date hereof and are subject to change without prior notification. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Other than certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst's judgment.

Goldman Sachs conducts a global full-service, integrated investment banking, investment management, and brokerage business. We have investment banking and other business relationships with a substantial percentage of the companies covered by Global Investment Research. Goldman Sachs & Co. LLC, the United States broker dealer, is a member of SIPC (https://www.sipc.org).

Our salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients and principal trading desks that reflect opinions that are contrary to the opinions expressed in this research. Our asset management area, principal trading desks and investing businesses may make investment decisions that are inconsistent with the recommendations or views expressed in this research.

The analysts named in this report may have from time to time discussed with our clients, including Goldman Sachs salespersons and traders, or may discuss in this report, trading strategies that reference catalysts or events that may have a near-term impact on the market price of the equity securities discussed in this report, which impact may be directionally counter to the analyst's published price target expectations for such stocks. Any such trading strategies are distinct from and do not affect the analyst's fundamental equity rating for such stocks, which rating reflects a stock's return potential relative to its coverage universe as described herein.

We and our affiliates, officers, directors, and employees will from time to time have long or short positions in, act as principal in, make a market or act as liquidity provider in, and buy or sell, any securities, derivatives, or other financial instruments referred to in this research (including any financial instruments issued by the companies referred to herein), unless otherwise prohibited by regulation or Goldman Sachs policy.

The views attributed to third party presenters at Goldman Sachs arranged conferences, including individuals from other parts of Goldman Sachs, do not necessarily reflect those of Global Investment Research and are not an official view of Goldman Sachs.

Any third party referenced herein, including any salespeople, traders and other professionals or members of their household, may have positions in the products mentioned that are inconsistent with the views expressed by analysts named in this report.

This research is not an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. It does not constitute a personal recommendation or take into account the particular investment objectives, financial situations, or needs of individual clients. Clients should consider whether any advice or recommendation in this research is suitable for their particular circumstances and, if appropriate, seek professional advice, including tax advice. The price and value of investments referred to in this research and the income from them may fluctuate. Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. Fluctuations in exchange rates could have adverse effects on the value or price of, or income derived from, certain investments.

Certain transactions, including those involving futures, options, and other derivatives, give rise to substantial risk and are not suitable for all investors. Investors should review current options and futures disclosure documents which are available from Goldman Sachs sales representatives or at https://www.theocc.com/about/publications/character-risks.jsp and https://www.goldmansachs.com/disclosures/cftc_fcm_disclosures. Transaction costs may be significant in option strategies calling for multiple purchase and sales of options such as spreads. Supporting documentation will be supplied upon request.

Differing Levels of Service provided by Global Investment Research: The level and types of services provided to you by Goldman Sachs Global Investment Research may vary as compared to that provided to internal and other external clients of GS, depending on various factors including your individual preferences as to the frequency and manner of receiving communication, your risk profile and investment focus and perspective (e.g.,

marketwide, sector specific, long term, short term), the size and scope of your overall client relationship with GS, and legal and regulatory constraints. As an example, certain clients may request to receive notifications when research on specific securities is published, and certain clients may request that specific data underlying analysts' fundamental analysis available on our internal client websites be delivered to them electronically through data feeds or otherwise. No change to an analyst's fundamental research views (e.g., ratings, price targets, or material changes to earnings estimates for equity securities), will be communicated to any client prior to inclusion of such information in a research report broadly disseminated through electronic publication to our internal client websites or through other means, as necessary, to all clients who are entitled to receive such reports.

All research reports are disseminated and available to all clients simultaneously through electronic publication to our internal client websites. Not all research content is redistributed to our clients or available to third-party aggregators, nor is Goldman Sachs responsible for the redistribution of our research by third party aggregators. For research, models or other data related to one or more securities, markets or asset classes (including related services) that may be available to you, please contact your GS representative or go to https://research.gs.com.

Disclosure information is also available at https://www.gs.com/research/hedge.html or from Research Compliance, 200 West Street, New York, NY 10282.

© 2026 Goldman Sachs.

You are permitted to store, display, analyze, modify, reformat, and print the information made available to you via this service only for your own use. You may not resell or reverse engineer this information to calculate or develop any index for disclosure and/or marketing or create any other derivative works or commercial product(s), data or offering(s) without the express written consent of Goldman Sachs. You are not permitted to publish, transmit, or otherwise reproduce this information, in whole or in part, in any format to any third party without the express written consent of Goldman Sachs. You may not use, extract, download or retrieve this information, in whole or in part, to train, finetune, adjust, influence the weights or behavior of, or otherwise develop or improve a machine learning or artificial intelligence model or system, or provide or reproduce this information, in whole or in part, as a prompt or input to any such system without the express written consent of Goldman Sachs. Furthermore, you may not bulk download this information, or employ any robot, spider, crawler, scraper, automated agent, or other electronic means to scrape, harvest, extract, or retrieve this information, without the express written consent of Goldman Sachs. A breach of the above terms may result in termination of your right to access this service.